

Python Interview Questions - Comprehensive Guide

1. Basic Python Interview Questions

- What are the key features of Python?
- Why is Python called an interpreted language?
- What is the difference between Python 2 and Python 3?
- What are Python's data types?
- What is the difference between a list, tuple, set, and dictionary?
- How is memory managed in Python?
- What is the difference between 'is' and '==' in Python?
- What are Python's numeric data types?
- What are mutable and immutable types in Python?
- How do you handle errors/exceptions in Python?

2. Intermediate Python Questions

- What is the difference between @staticmethod and @classmethod?
- What are *args and **kwargs in Python?
- What is list comprehension? Give an example.
- Explain lambda functions with an example.
- What is the difference between append() and extend() in lists?
- What are Python decorators? When do you use them?
- How do you handle files in Python? (Explain with open())
- What is the use of enumerate()?
- How does Python's garbage collector work?
- What are iterators and generators?

3. Advanced Python Interview Questions

- How is memory allocated to variables in Python?

- What are Python namespaces and scopes?
- Explain the GIL (Global Interpreter Lock) in Python.
- How do you optimize performance in Python applications?
- What are metaclasses in Python?
- What are the differences between shallow and deep copy?
- What is monkey patching in Python?
- What is the difference between threading and multiprocessing?
- How do you serialize data in Python? (e.g., with pickle, json)
- What are context managers?

4. OOPs in Python

- What is the difference between class and object?
- How do you achieve encapsulation, inheritance, and polymorphism in Python?
- What is multiple inheritance? How is it handled in Python?
- Explain the concept of MRO (Method Resolution Order).
- What is the difference between `__init__` and `__new__`?
- How do private, protected, and public members work in Python?
- How to implement abstraction in Python?

5. Python with Real-Time Scenarios

- How did you use Python in your recent project?
- In which scenarios did Python help you reduce development time?
- How do you handle data preprocessing in Python?
- Can you explain how you used Pandas for data analysis?
- How did you connect Python with a database (SQLite/MySQL)?
- How did you build the GUI for your Python project?
- Did you use Python for web scraping or automation?
- How did you handle performance issues in your Python script?

- What testing methods do you use in Python?
- How did you manage large data files (CSV, Excel) in Python?

6. Data Structures and Algorithms in Python

- How do you implement a stack and queue in Python?
- How does Python's dictionary handle collisions?
- Write a program to find duplicates in a list.
- Write a Python function to reverse a string.
- How do you implement binary search in Python?
- What is the difference between recursion and iteration?
- How do you sort a list of dictionaries by a key?
- How would you detect a cycle in a graph using Python?
- Explain time complexity with a real example in Python.
- Can you write a program to find the factorial of a number?

7. Database & Integration Questions (Python with SQL)

- How do you connect Python with MySQL/MongoDB?
- How do you execute queries using Python?
- What is the difference between cursor and connection objects?
- How do you handle transactions and rollbacks in Python?
- How did you handle exceptions in database access?

8. Python Libraries (for Real-Time Use)

- How did you use Pandas for data analysis?
- What is the use of NumPy in your project?
- How did you visualize data using Matplotlib or Seaborn?
- What is the difference between loc and iloc in Pandas?
- Have you used Flask or Django? What's the difference?
- How did you use OpenCV in your stress detection project?

9. Behavioral / HR + Client-Understanding Questions

- Why did you choose Python for your project?
- How do you keep yourself updated with new features in Python?
- Have you ever solved a production bug in Python? How?
- What challenges did you face while using Python in real projects?
- Can you explain a time when Python helped automate a task?